Rachel Baker-Ramos

Computer Science Ph.D. Student

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Research Interests

Educational Technology: Culturally-Relevant STEM and SEL Education, Narrative/Game-Based Learning, AI for Low-Resource Language Education. **Community-Based Co-Design & Indigenous HCI:** Data Sovereignty, Data Storytelling, Traditional Knowledge Systems, Speculative Design & Futuring.

Education

2026– **Ph.D. Computer Science**, *Georgia Institute of Technology*
Advisors: Josiah Hester (Georgia Tech), Alex Cabral (M.I.T.).
- **Stamps Doctoral Fellowship**

2022–24 **M.S. Human-Computer Interaction**, *Georgia Institute of Technology*
Thesis: Co-designing Culturally-Relevant Generative EdTech with Hawaiian Immersion Educators, Advisors: Josiah Hester, Judith Uchidiuno

2014–19 **Bachelor of Architecture (BArch)**, *Carnegie Mellon University*
Additional major: Social & Political History. Graduated with University Honors and College of Fine Arts Honors.

Publications

Conference Papers

CHI 2027 **Rachel Baker-Ramos**, Candace Boice, Josiah Hester, Alex Cabral. "I Believe Candidate G Has Ulterior Motives:" Operationalizing Narrative Policy Framework to Study Data Center Narratives in US Elections. ACM CHI Conference on Human Factors in Computing Systems. (Under Review)

SIGCSE TS 2027 **Rachel Baker-Ramos**, Evyn-Bree Helekahi-Kaiwi, Alakaʻi Iaea-Russell, Lokelani Chanelle Tolentino, Rebecca Diego, Josiah Hester. AI4Hawaiʻi: Culturally-Grounded AI Literacy Training for K-12 Hawaiian Education Teachers. ACM SIGCSE Technical Symposium 2027. (Accepted)

IDC 2026 * **Rachel Baker-Ramos**, Manas Mhasakar, Viswak Raja, Evyn-Bree Helekahi-Kaiwi, William Gelder, Benjamin Carter, Rebecca Diego, Alex Cabral, Josiah Hester. Kumu Connect: Culturally-Authentic AI Tools to Enable Educator Agency and Data Sovereignty for Hawaiian K-12 CS Education. 25th Annual ACM Interaction Design and Children Conference. (Acceptance rate: 23%)
Best Paper Honorable Mention

CHI 2026 * Julia Aileen McKenna, Gabriela Buraglia, Jahanvi Kolakaluri, **Rachel Baker-Ramos**, Samantha Carter, Joe Graveen, Jonathan Gilbert, James Rasmussen, Brandon Byrne, Darren Vogt, Josiah Hester, Kim Marion Suiseeya, Alex Cabral. Noondawind: Co-Designed Dashboard for Indigenous Data Access and Environmental Policy Implementation. ACM CHI Conference on Human Factors in Computing Systems.
Best Paper Honorable Mention

CHI 2026 Dora Zhao, Hannah Cha, Michael J Ryan, Angelina Wang, **Rachel Baker-Ramos**, Evyn-Bree Helekahi-Kaiwi, Rebecca Diego, Josiah Hester, Diyi Yang. Whose Knowledge Counts? Co-Designing Community-Centered AI Auditing Tools with Educators in Hawaiʻi. ACM CHI Conference on Human Factors in Computing Systems.

RESPECT 2025 **Rachel Baker-Ramos**, Will Gelder, Leah Cho, Jahnavi Kolakaluri, Josiah Hester. Kumu Connect: Design Thinking for Place-Based Generative Educational Technology in Hawaiian Immersion Schools. 2025 ACM Conference on Research on Equitable and Sustained Participation in Engineering, Computing, and Technology.

RESPECT 2024 William Gelder, **Rachel Baker-Ramos**, Ayoung Cho, Jahnavi Kolakaluri, Judith Uchidiuno, Josiah Hester. "Those don't work for us": An Assets-Based Approach to Incorporating Emerging Technologies in Viable Hawaiian Teacher Support Tools for Culturally Relevant CS Education. 2024 ACM Conference on Research on Equitable and Sustained Participation in Engineering, Computing, and Technology.

Koli Calling 2023 Michael J. Johnson, **Rachel Baker-Ramos**, Christopher L. Hovey, Betsy DiSalvo. Keeping Mindful of Modality: A Comparison of Computer Science Education Resources for Learning. 23rd ACM Koli Calling International Conference on Computing Education Research.)

Posters and Abstracts

SIGCSE TS 2027 **Rachel Baker-Ramos**. Through The Stories We Tell: Grounding STEM Educational Technology in Indigenous Epistemologies. ACM SIGCSE Technical Symposium 2027. (Under Review)

GoodIT 2026 **Rachel Baker-Ramos**, Josiah Hester, Alex Cabral. Still Warm: A Digital Game for Players to Address Climate Grief. ACM 6th International Conference on Information Technology for Social Good. (Accepted)

CHI 2025 **Rachel Baker-Ramos***, Manas Mhasakar*, Ben Carter, Evyn-Bree Helekahi-Kaiwi, Josiah Hester. "I Would Never Trust Anything Western": Kumu (Educator) Perspectives on Use of LLMs for Culturally Revitalizing CS Education in Hawaiian Schools. ACM CHI Conference on Human Factors in Computing Systems. *equal contribution

CHI PLAY 2024 **Rachel Baker-Ramos**, Michael Parkin, Angelica Raza-Furtado, Alika S. Naihe, Jack Hobbs, Jeffrey Vierra, Aarom Ramirez, Chantel Rubin-Molina, Cheradean Kaaialii, Celeste Mason, Yaman Sangar, May Okihiro, Josiah Hester. Enabling Restorative Hawaiian Futurism via a Tabletop Role-Playing Game and Digital Haʻi Moʻolelo (Storytelling). 2024 ACM Annual Symposium on Computer-Human Interaction in Play.

Awards & Fellowships

2026 **Stamps Doctoral Fellowship**, *Georgia Institute of Technology*
Georgia Tech's most prestigious merit-based doctoral fellowship (1 of 9 selected from 4,300+ applicants), created to attract exceptional scholars and support their development into leaders who will advance technology and improve the human condition.

2026 **Presidential Fellowship**, *Georgia Institute of Technology*
Offered upon admission; replaced by the Stamps Fellowship prior to enrollment.

2026 **Research Administration & Support Spotlight Award**, *Georgia Institute of Technology*

2025 **Outstanding Research Scientist Award**, *Georgia Institute of Technology, College of Computing*

2024 **Best MS-HCI Thesis Award Finalist**, *Georgia Institute of Technology, MS-HCI Program*

2023 **Civic Innovation Fellowship**, *Coding It Forward*
Selected from 2,500 student applicants (5% acceptance rate) for the Coding It Forward Summer Fellowship, a program for early-career technologists to explore civic & public interest technology.

Work Experience

2026– **Graduate Assistant**, *College of Computing, Georgia Institute of Technology*
Mentor 4 MS/BS students in their independent studies and theses. Provide support for Ka Moamoa Lab projects (research protocol review, co-design and community-based design advising, IRB protocol support, paper writing & graphics). Write and prepare grant proposals.

2024–26 **Research Scientist**, *College of Computing, Georgia Institute of Technology*
Led the Ka Moamoa Lab's education & public interest HCI sub-group, conducting user research, authoring 7 peer-reviewed publications, managing 13 MS/BS students across 7+ projects. Wrote and prepared grant and award proposals, securing 475k funding from Google Research, Mozilla Foundation, and the NSF.

2023–24 **Graduate Research Assistant**, *College of Computing, Georgia Institute of Technology*

2023 **UX Design Fellow**, *U.S.C.I.S. Refugee, Asylum & International Operations Directorate*

2019–22 **Project Designer**, *Caddis Collaborative* (*Architecture & Urban Design Studio*)

Service & Volunteering

2024– **UX Design & Research Advisor**, *Ulu Lāhui Foundation*

June 2026 **ACM Interaction Design and Children Conference**, *Student Volunteer*

2024 **UX of EdTech**, *Community Volunteer, Summer Cohort*

2023–24 **MS-HCI DEI Forum**, *DEI Forum Co-leader*

2020–23 **BoulderReads**, *Volunteer ESL Literacy Tutor*

2020–22 **Lead Graphic Designer**, *Veggie Mijas* (*Latina/x/e Environmental Justice Org*)

Grantwriting & Funding

2026 **Mozilla Responsible Computing Challenge**, *Mozilla Foundation*
Project: Footsteps Before Ours: An Indigenous-Driven Responsible Computing Pathway Rooted in Relational Accountability, Role: Ph.D. Student Researcher & Grant-writer

2026 **MIT SA+P HASS Award**, *MIT School of Architecture and Planning* (*SA+P*)
Project: Still Warm: A Narrative-driven Game on Climate Grief, Role: Ph.D. Student Researcher & Grant-writer

2026 **NSF Supplement Support Request: Translating Community-Driven Edge AI Data Systems into K–12 STEM Curricula for Rural Classrooms**, *National Science Foundation*
Project: SCC-IRG: Edge AI, Biofabrication, and Data Science for Disaster Preparedness and Community Wellbeing in Hawaii, Role: Personnel & Grant-writer

2025 **Indigenous Communities Fellowship**, *MIT Solve*
Project: Kumu Connect, Role: Senior Personnel & Grant-writer

2025 **Rural & Indigenous Community Catalyst Program**, *aiEDU*
Project: AI for Hawai'i (AI4HI) Curricula Co-Creation Workshops & Resource Hub, Role: Senior Personnel & Grant-writer

2024 **Google Academic Research Award**, *Google Research*
Project: Kumu Connect: Community-informed AI Teacher Support Tools for Place-Based CS Curriculum in Hawaiian Schools, Role: Senior Personnel & Grant-writer